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- Find n^{th} column title

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Addition & Multiplication Rule

01. Given 3 True/False questions, every question has to be answered. In how many ways can you answer all the questions.

$$\underbrace{2} \times \underbrace{2} \times \underbrace{2} = 2^3 \Rightarrow 8 \text{ ways}$$

Diagram illustrating the multiplication rule for 3 questions, each with 2 possible answers (T or F):

- Question 1: T, F
- Question 2: T, F
- Question 3: T, F

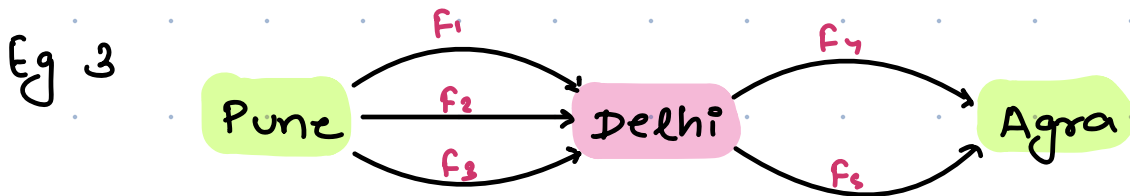
F	F	F
F	F	T
F	T	F
F	T	T
T	F	F
T	T	F
T	F	T
T	T	T

8 ways

Q2 Given 10 girls & 7 boys. How many different pairs can we form?

Note :- pair $\rightarrow$ 1 Girl + 1 Boy

B_1	G_1		
B_2	G_2	1 Boy	1 Girl
B_3	G_3	7	$\times$ 10 = 70 ways
B_4	G_4		
B_5	G_5		
B_6	G_6		
B_7	G_7		
	$\vdots$		
	G_{10}		



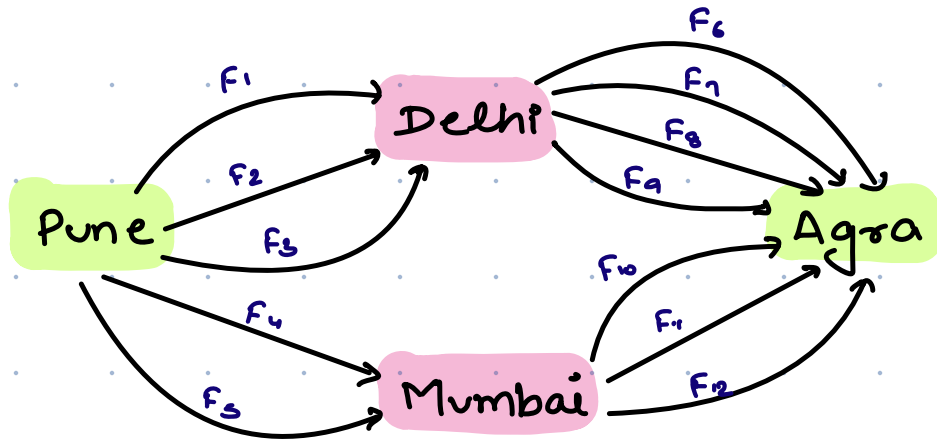
No. of ways to reach Agra from Pune via Delhi

Pune $\rightarrow$ Delhi AND Delhi $\rightarrow$ Agra

3 $\times$ 2 = 6 ways

$F_1 F_4$ $F_2 F_4$ $F_3 F_4$ $F_1 F_5$ $F_2 F_5$ $F_3 F_5$

Eg 4



No. of ways to reach Agra from pune?

via Delhi

OR

via Mumbai

Pune $\rightarrow$ Delhi AND Delhi $\rightarrow$ AgraP $\rightarrow$ M AND M $\rightarrow$ A $3 * 4$ $2 * 3$ $\Rightarrow 12$

+

6

 $\rightarrow 18 \text{ ways}$ OR $\rightarrow +$ AND $\rightarrow *$

* Permutation

→ Arrangement of objects

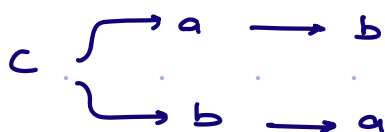
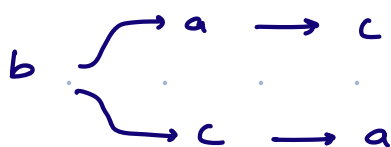
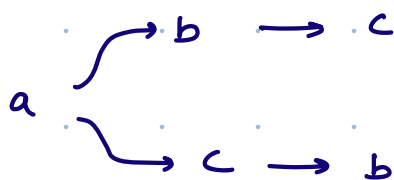
→ In permutation, order matter

$$(i, j) \neq (j, i)$$

Q Given 3 distinct characters, In how many ways can you arrange them.

str = "a b c"

$$\underline{3} * \underline{2} * \underline{1} = 3! = 6 \text{ ways}$$



* In how many ways n distinct characters can be arranged.

$$\underline{N} * \underline{N-1} * \underline{N-2} * \underline{(N-3)} * \dots * \underline{1} \Rightarrow \underline{\underline{N!}}$$

Q Given 5 distinct characters, in how many ways can we arrange 2 characters

str="a b c d e"

$$\underline{5} * \underline{4} = \underline{\underline{20 \text{ ways}}}$$

a → b, c, d, e

b → a, c, d, e

c → a, b, d, e

d → a, b, c, e

e → a, b, c, d

* Given N distinct characters, in how many ways we can arrange 3 characters

$$\underline{N} * \underline{(N-1)} * \underline{(N-2)} = \underline{N(N-1)(N-2) \text{ ways}}$$

* Given N distinct characters, in how many ways we can arrange r characters

$$N * (N-1) * (N-2) * \dots * (N-r+1) * (N-r) * (N-r-1) * \dots * 1$$

$$(N-r) * (N-r-1) * \dots * 1$$

$$\Rightarrow \frac{N!}{(N-r)!} = {}^N P_r \Rightarrow \text{No. of ways to arrange } r \text{ place from } N \text{ distinct characters}$$

* Combination

↳ No. of ways to select something

↳ Order of selection doesn't matter

$$(i, j) = (j, i)$$

Q Given 4 players, count the no. of ways to select 3 players.

$$\{ P_1, P_2, P_3, P_4 \}$$

P_1, P_2, P_3

P_1, P_2, P_4

P_1, P_3, P_4

P_2, P_3, P_4

} 4 combination

<u>P₁</u>	<u>P₂</u>	<u>P₃</u>
<u>P₁</u>	<u>P₃</u>	<u>P₂</u>
<u>P₂</u>	<u>P₁</u>	<u>P₃</u>
<u>P₂</u>	<u>P₃</u>	<u>P₁</u>
<u>P₃</u>	<u>P₁</u>	<u>P₂</u>
<u>P₃</u>	<u>P₂</u>	<u>P₁</u>

P₁ P₂ P₃

<u>P₁</u>	<u>P₂</u>	<u>P₄</u>
<u>P₁</u>	<u>P₄</u>	<u>P₂</u>
<u>P₂</u>	<u>P₁</u>	<u>P₄</u>
<u>P₂</u>	<u>P₄</u>	<u>P₁</u>
<u>P₄</u>	<u>P₁</u>	<u>P₂</u>
<u>P₄</u>	<u>P₂</u>	<u>P₁</u>

P₁ P₂ P₄

<u>P₁</u>	<u>P₃</u>	<u>P₄</u>
<u>P₁</u>	<u>P₄</u>	<u>P₃</u>
<u>P₃</u>	<u>P₁</u>	<u>P₄</u>
<u>P₃</u>	<u>P₄</u>	<u>P₁</u>
<u>P₄</u>	<u>P₁</u>	<u>P₃</u>
<u>P₄</u>	<u>P₃</u>	<u>P₁</u>

P₁ P₃ P₄

<u>P₂</u>	<u>P₃</u>	<u>P₄</u>
<u>P₂</u>	<u>P₄</u>	<u>P₃</u>
<u>P₃</u>	<u>P₂</u>	<u>P₄</u>
<u>P₃</u>	<u>P₄</u>	<u>P₂</u>
<u>P₄</u>	<u>P₂</u>	<u>P₃</u>
<u>P₄</u>	<u>P₃</u>	<u>P₂</u>

P₂ P₃ P₄

For every selection of 3 players = 3! = 6 arrangements

Total no. of arrangements = No. of arrangements * No. of selection
for 1 selection

$$24 = 6 * x$$

$$x = \frac{24}{6} = 4$$

Q Given n distinct ele, in how many ways can we go & select r elements

$$\text{Total no. of arrangements} = {}^n P_r = \frac{n!}{(n-r)!}$$

No. of arrangement for one selection $\Rightarrow r!$
(pick r elements)

$$\text{No. of selections} \times r! = \frac{n!}{(n-r)!}$$

$$\text{No. of selection} = \frac{n!}{r! (n-r)!}$$

$${}^n C_r = \frac{n!}{r! (n-r)!}$$

* Properties of Combination

01. No. of ways to select 0 items out of N items

$${}^N C_0 = \frac{N!}{0! (N-0)!} = \frac{\cancel{N!}}{1 \times \cancel{N!}} \Rightarrow 1$$

02. No. of ways to select N items out of N items

$${}^N C_N = \frac{N!}{N! (N-N)!} = \frac{\cancel{N!}}{\cancel{N!} * 0!} = 1$$

03. No. of ways to select r items out of N items

$${}^N C_r = \frac{N!}{r! (N-r)!}$$

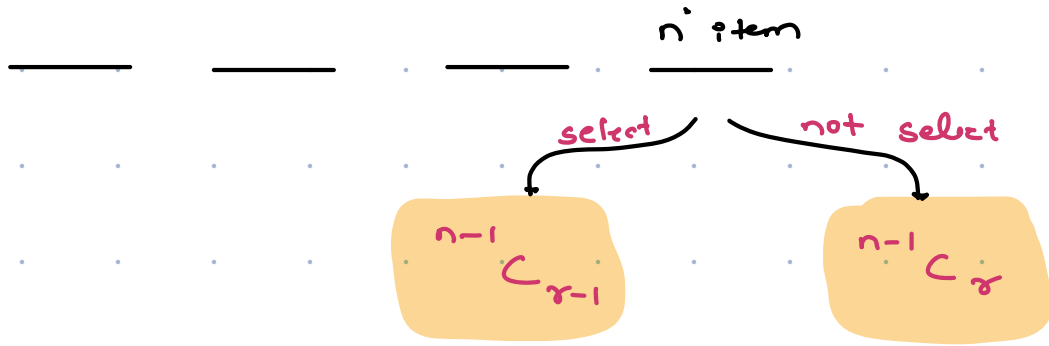
04. No. of ways to select $(N-r)$ items out of N items

$${}^N C_{N-r} = \frac{N!}{(N-r)! * (N-(N-r))!}$$

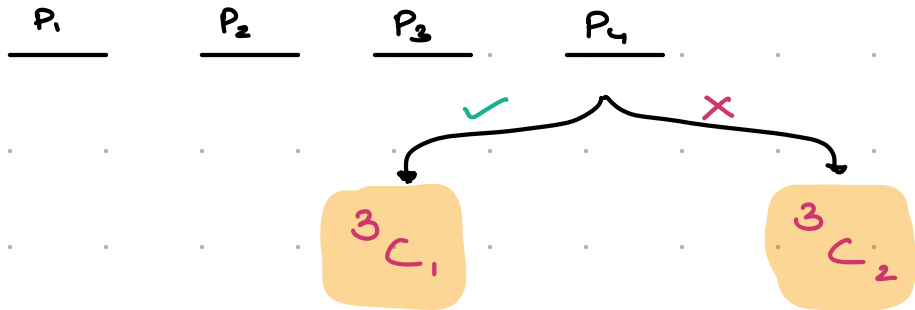
$$= \frac{N!}{(N-r)! * r!}$$

$${}^N C_r = {}^N C_{N-r}$$

* Given N distinct elements, select r elements



4 players & we have to select 2



$${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$$

* Pascal Triangle

Given n , generate pascal triangle for this no. n

$$\underline{\underline{N=4}}$$

	0	1	2	3	4	
0	1					0C_0
1	1	1				1C_0 1C_1
2	1	2	1			2C_0 2C_1 2C_2
3	1	3	3	1		3C_0 3C_1 3C_2 3C_3
4	1	4	6	4	1	4C_0 4C_1 4C_2 4C_3 4C_4

Brute force → For each & every position, we will generate a value through nC_r & print it.

$${}^nC_r = \frac{N!}{r! (N-r)!}$$

$$5! = 120$$

$$10! = 3\,628\,800$$

$$20! = 2.4 \times 10^{18}$$

} factorials grow very rapidly

	0	1	2	3	4
0	1				
1	1	1			
2	1	2	1		
3	1	3	3	1	
4	1	4	6	4	1

$$* {}^nC_0 = 1$$

$$* {}^nC_n = 1$$

$$* {}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$$

$${}^4C_1 = {}^3C_0 + {}^3C_1$$

$${}^iC_j = {}^{i-1}C_{j-1} + {}^{i-1}C_j$$

```
int [][] c = new int[n+1][n+1]
```

```
System.out.println("1"); // O*row
```

```
for (i=1 ; i<=n ; i++) {
```

```
    c[i][0] = 1
```

```
    c[i][i] = 1
```

```
    System.out.print(c[i][0]);
```

```
    for (j=1 ; j<i ; j++) {
```

```
        c[i][j] = c[i-1][j-1] + c[i-1][j]
```

```
        System.out.print(c[i][j]);
```

```
    }
    System.out.print(c[i][i]);
```

TC : $O(n^2)$

SC : $O(n^2)$

* Find the n^{th} column title, where columns are titled as

1 2 3 4 26 27 28 52 53
 A B C D Z AA AB AZ BA BB ... BZ ... ZZ AAA

$n = 3 \longrightarrow \text{Ans} = C$

$n = 27 \longrightarrow \text{Ans} = AA$

$n = 30 \longrightarrow \text{Ans} = AD$

$n = 50 \longrightarrow \text{Ans} = AX$

Decimal no.

0 10 20 30

1 11 21 31

2 12 32

3 13

4

.

5

.

6

.

.

.

9

19

29

39

Column Title

A AA BA

B AB BB

C AC BC

D .

E .

.

.

.

Y AY BY

Z AZ BZ

Map

1 → A
2 → B
3 → C
4 → D
⋮
26 → Z

N = 50

26	50	
26	1	24 → X
	0	1 → A

Ans = AX

N = 30

26	30	
26	1	4 → D
	0	1 → A

Ans = AD

N = 100

26	100	
26	3	22 → V
	0	3 → C

Ans = CV

$$52 = 26 * 2 + 0$$

$$52 = 26 * 1 + 26$$

26	52	
26	2	0 →
	0	2 → B

Nothing mapped with 0

Ans = AZ

26	52	
26	2 1	0 26 → Z
	0	1 → A

Ans = AZ

26	78	
26	3 2	0 26 → Z
	0	2 → B

$$78 = 26 * 3 + 0$$

$$78 = 26 * 2 + 26$$

Ans = BZ

* Create a map with all mappings

String ans = ""

while (n > 0) {

int rem = n % 26;

n = n / 26;

if (rem == 0) {

rem = 26

n = n - 1

}

ans = map.get(rem) + ans;

}

return ans

$$\left. \begin{array}{l} 1 \rightarrow A \\ 2 \rightarrow B \\ 3 \rightarrow C \\ 4 \rightarrow D \\ \vdots \\ 26 \rightarrow Z \end{array} \right\}$$

26	1000	
26	38	12
26	1	12
	0	1

Ans = ALL

* HashMap

```
HashMap < Integer, Character > map = new HashMap<>();
```

```
for (int i=1; i ≤ 26; i++) {
```

```
    char ch = (char)(i + 'A' - 1);
```

```
    map.put(i, ch);
```

```
}
```